

Worksheet 102 · Spot on facts, pp. 226–227

Technologies that change society

The key facts at a glance — the connections are yours to make. Your basis for the Kolloquium.

1 Developments in the 21st century

Genetic engineering

CRISPR edits the genetic code precisely.

+ drought-resistant crops, new treatments

- morality; *interference* with DNA

Eco-technology

Solar, hydrogen and wind power are cheaper; more e-cars.

+ less fossil fuel, ocean clean-ups

- raw materials (lithium) harm the environment

AI & digital content

Prompts to large language models create texts, media, programs.

+ information re-used in myriad contexts

- where does content come from — can we trust it?

Connect: All three bring a hope and a new problem. Which development shows this most clearly — and why?

2 Artificial intelligence (AI)

AI = *the simulation of human intelligence in machines*: learning, problem-solving, *reasoning*, perceiving.

Weak / narrow AI

single tasks, e.g. personal assistants, video games

Strong / general AI

complex, human-like, e.g. (book) self-driving cars, robots*

Impact

+ everyday life, education, health

- jobs, deepfakes, misinformation, copyright

* Most experts today call self-driving cars *narrow AI*; *general AI* means a still hypothetical human-level intelligence.

Connect: Why does checking data matter more than ever now that AI creates content?

3 Augmented reality (AR) and virtual reality (VR) — both extended reality (XR)

AR

What it does adds digital *overlays* to the real world

Devices phone, tablet (camera)

Used in entertainment, medicine, shopping

VR

replaces the real world with a 360° environment

headset, motion controllers

gaming, tourism, education, medicine, *warfare*

Connect: How are AR, VR and AI related? Name one example you know (e.g. *Virtual Helsinki*, p. 230).

4 Utopia or dystopia — where will technology take us?

Utopia *ou-topos* = “no place”, *eu-topos* = “good place” (Thomas More, 1516).

An **ideal society** that criticises the present by showing a better world.

e.g. Butler, *Erewhon* (1872); Bellamy, *Looking Backward* (1888)

Dystopia *dys-topos* = “bad place”.

A **worst-case society**, often in the near future, that warns about *contemporary* problems.

e.g. *Brave New World* (1932), *Nineteen Eighty-Four* (1949), *The Hunger Games* (2008)

Connect: Which technology from boxes 1–3 could be the starting point of a dystopia? Describe that society in one or two sentences.

Worksheet 102 · Task

Kolloquium Carousel

In Part 2 of the Kolloquium you answer questions on topics from other semesters — no notes. Train it now.

How it works

- 1 Double circle: inside **candidates**, outside **examiners**.
- 2 Examiner reads a card — 15 s to think, **2 min** to answer.
- 3 Follow-up question (1 min), then rubric + two stars and a wish.
- 4 Outer circle moves on. Swap roles after two rounds.

Answer like a candidate

Define



Explain



Example



Evaluate

Question cards (follow-up in italics)

1 AR vs VR

Explain the difference between AR and VR.

Follow-up: Which will change everyday life more?

2 AI

What is AI? Weak vs strong AI.

Follow-up: Where does AI help — where does it worry you?

3 Trust

Why is it harder to know where online content comes from?

Follow-up: How can you protect yourself?

4 Genes

What does CRISPR make possible?

Follow-up: Where would you draw the line?

5 Terms

Where do *utopia* and *dystopia* come from?*Follow-up: Why are dystopias so popular today?*

6 Big picture

Which technology is the most life-changing?

Follow-up: How will it affect your children's lives?

Examiner's sheet 0 = not yet · 1 = partly · 2 = mostly · 3 = fully

	Round 1	Round 2
Candidate		
Content — accurate facts and terms	/3	/3
Structure — clear line of thought	/3	/3
Language — topic words, range, accuracy	/3	/3
Interaction — reacts to the follow-up	/3	/3
Total	/12	/12

After the carousel — your Kolloquium note Which card was hardest? What would you need to answer it well next time?
